



HK IMO Training 2022-2023
Problem Set 6
For 5 November 2022



Problem 21. Find all monic polynomials f with integer coefficients such that p divides $2(f(p)!) + 1$ for all sufficiently large primes p for which $f(p)$ is a positive integer.

Problem 22. Let m and n be odd positive integers. We colour each cell of an $m \times n$ table black or white. Let x be the number of rows with more black cells than white cells, and let y be the number of columns with more white cells than black cells. Find the largest possible value of $x + y$.

Problem 23. Let a, b, c, d be real numbers such that $|a|, |b|, |c|, |d| > 1$ and

$$a + b + c + d + abc + abd + acd + bcd = 0.$$

Prove that

$$\frac{1}{a-1} + \frac{1}{b-1} + \frac{1}{c-1} + \frac{1}{d-1} > 0.$$

Problem 24. Let $ABCD$ be a quadrilateral inscribed in a circle ω with centre O . Let P be the intersection of two diagonals AC and BD . Let Q be a point lying on the segment OP . Let E and F be the orthogonal projections of Q on the lines AD and BC , respectively. The points M and N lie on the circumcircle of triangle QEF such that $QM \parallel AC$ and $QN \parallel BD$. Prove that the two lines ME and NF meet on the perpendicular bisector of segment CD .